

CEN4025C Module 1 Project

This is a walkthrough of the Angular environment setup. We will install Angular and Material, the Angular UI framework. We will add a simple row of dummy buttons, and a basic list of Pizza Restaurant customers.

Install angular using npm. Note that the latest version is 17.x, versus our textbook which is using 13, so some commands will change.

```
D:\fscj\CEN4025C-SoftwareDevII\cen4025c-Summer24>npm install -g
@angular/cli
added 236 packages in 15s
44 packages are looking for funding
  run `npm fund` for details
npm notice
npm notice New minor version of npm available! 10.5.0 -> 10.7.0
npm notice Changelog: https://github.com/npm/cli/releases/tag/v10.7.0
npm notice Run npm install -g npm@10.7.0 to update!
npm notice
```

ng is the shortcut for running the Angular CLI

```
D:\fscj\CEN4025C-SoftwareDevII\cen4025c-Summer24>ng version
? Would you like to share pseudonymous usage data about this project with the Angular Team at
Google under Google's Privacy Policy at https://policies.google.com/privacy. For more details
and how to change this setting, see https://angular.io/analytics. No
Global setting: disabled
Local setting: No local workspace configuration file.
Effective status: disabled
```

```

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```

```
Angular CLI: 17.3.6
Node: 20.12.0
Package Manager: npm 10.5.0
OS: win32 x64
```

Angular:

...

Package	Version
@angular-devkit/architect	0.1703.6 (cli-only)
@angular-devkit/core	17.3.6 (cli-only)
@angular-devkit/schematics	17.3.6 (cli-only)
@schematics/angular	17.3.6 (cli-only)

Change your folder to the GitHub repo and create the Angular project (note that this is residing side-by-side with the Spring Boot project)

```
D:\fscj\CEN4025C-SoftwareDevII\cen4025c-Summer24>cd M1ProjectSolution
D:\fscj\CEN4025C-SoftwareDevII\cen4025c-Summer24\M1ProjectSolution>dir
Volume in drive D is Data
Volume Serial Number is FEA9-B58E
```

Directory of D:\fscj\CEN4025C-SoftwareDevII\cen4025c-Summer24\M1ProjectSolution

```
05/07/2024 08:19 PM <DIR>      .
05/07/2024 06:43 PM <DIR>      ..
05/07/2024 06:43 PM <DIR>      .github
05/07/2024 08:35 PM <DIR>      PizzaRestaurant
05/07/2024 08:04 PM          21 README.md
                1 File(s)      21 bytes
                4 Dir(s) 354,634,104,832 bytes free
```

Create the new project:

```
D:\fscj\CEN4025C-SoftwareDevII\cen4025c-Summer24\M1ProjectSolution>ng
new PizzaRestaurantFrontend
```

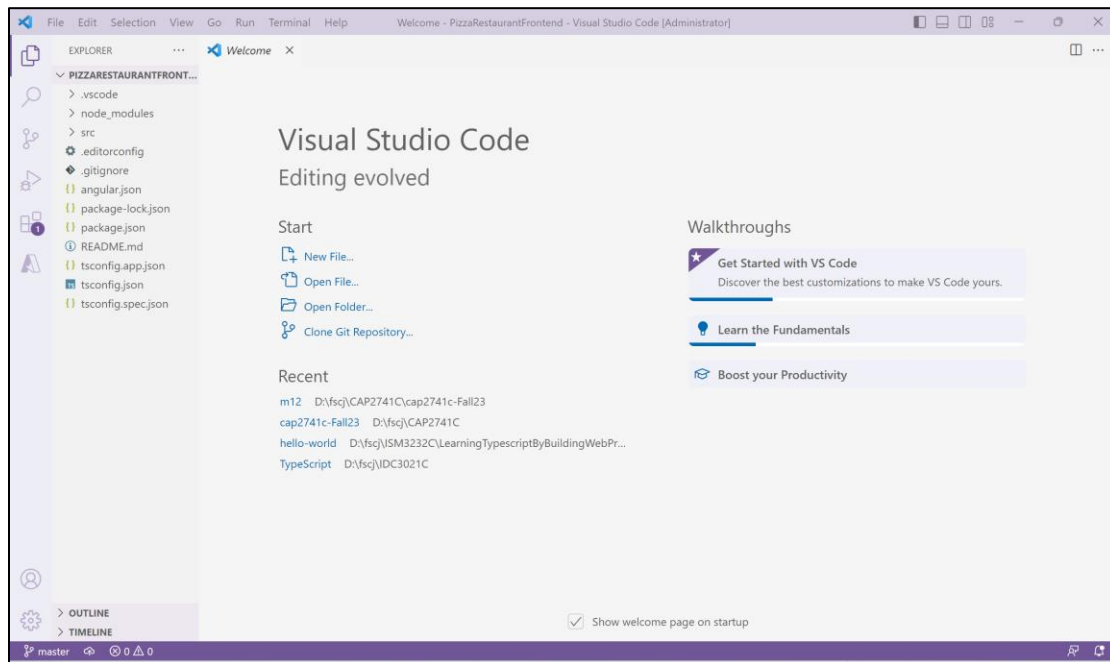
Use your cursor keys to select Sass (SCSS) from the options provided

```
? Which stylesheet format would you like to use? Sass (SCSS) [ https://sass-
lang.com/documentation/syntax#scss ]
? Do you want to enable Server-Side Rendering (SSR) and Static Site Generation
(SSG/Prerendering)? No
CREATE PizzaRestaurantFrontend/angular.json (2954 bytes)
CREATE PizzaRestaurantFrontend/package.json (1094 bytes)
CREATE PizzaRestaurantFrontend/README.md (1104 bytes)
CREATE PizzaRestaurantFrontend/tsconfig.json (889 bytes)
CREATE PizzaRestaurantFrontend/.editorconfig (290 bytes)
CREATE PizzaRestaurantFrontend/.gitignore (590 bytes)
CREATE PizzaRestaurantFrontend/tsconfig.app.json (277 bytes)
CREATE PizzaRestaurantFrontend/tsconfig.spec.json (287 bytes)
CREATE PizzaRestaurantFrontend/.vscode/extensions.json (134 bytes)
CREATE PizzaRestaurantFrontend/.vscode/launch.json (490 bytes)
CREATE PizzaRestaurantFrontend/.vscode/tasks.json (980 bytes)
```

CREATE PizzaRestaurantFrontend/src/main.ts (256 bytes)
CREATE PizzaRestaurantFrontend/src/favicon.ico (15086 bytes)
CREATE PizzaRestaurantFrontend/src/index.html (322 bytes)
CREATE PizzaRestaurantFrontend/src/styles.scss (81 bytes)
CREATE PizzaRestaurantFrontend/src/app/app.component.html (20239 bytes)
CREATE PizzaRestaurantFrontend/src/app/app.component.spec.ts (996 bytes)
CREATE PizzaRestaurantFrontend/src/app/app.component.ts (333 bytes)
CREATE PizzaRestaurantFrontend/src/app/app.component.scss (0 bytes)
CREATE PizzaRestaurantFrontend/src/app/app.config.ts (235 bytes)
CREATE PizzaRestaurantFrontend/src/app/app.routes.ts (80 bytes)
CREATE PizzaRestaurantFrontend/src/assets/.gitkeep (0 bytes)
✓ Packages installed successfully.
Successfully initialized git.

Move to the project folder and start VSCode

```
D:\fscj\CEN4025C-SoftwareDevII\cen4025c-Summer24\M1ProjectSolution>cd  
PizzaRestaurantFrontend  
D:\fscj\CEN4025C-SoftwareDevII\cen4025c-  
Summer24\M1ProjectSolution\PizzaRestaurantFrontend>code .
```



Start a terminal, then build and run the server:

ng serve

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL

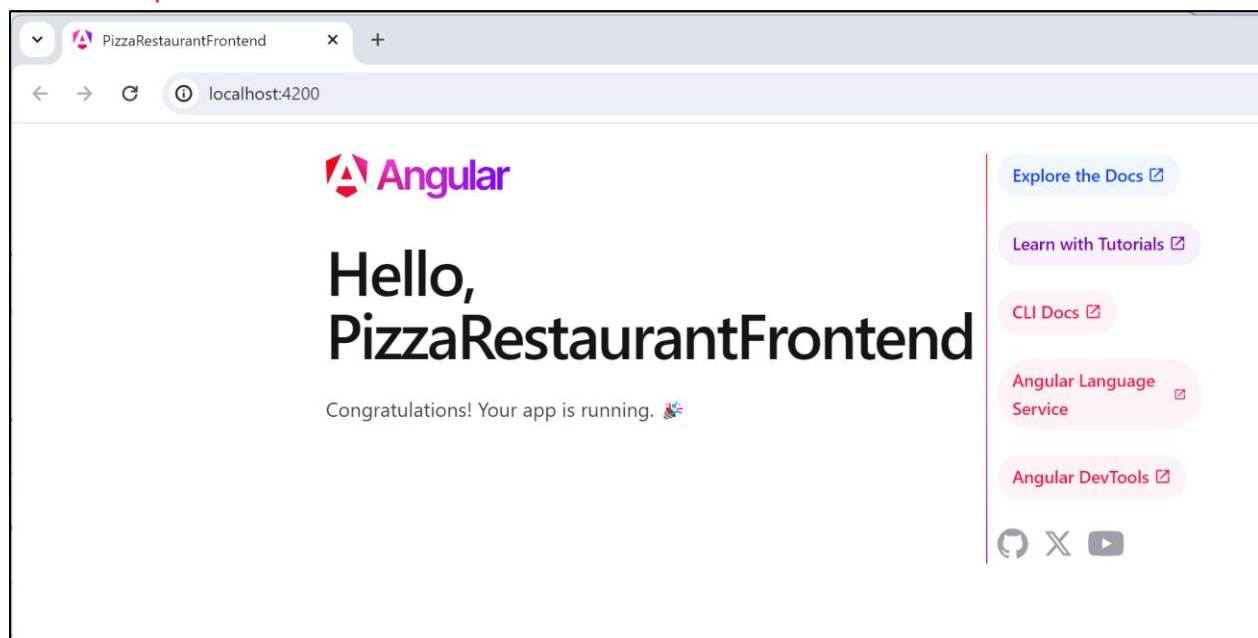
polyfills.js      | polyfills      | 85.81 kB |
main.js           | main           | 21.99 kB |
styles.css        | styles         | 96 bytes |

| Initial total | 107.89 kB

Application bundle generation complete. [1.574 seconds]

Watch mode enabled. Watching for file changes...
→ Local:  http://localhost:4200/
→ press h + enter to show help
```

Use the link provided to test the install



Ctrl-C to break out of the server in the VS Code Terminal. You can also take advantage of the watch feature; changes made in the project are automatically pushed ("hot reloaded") out to the running server without having to stop/rebuild/start.

Install the Materials UI framework in your project and select the various associated options:

```
ng add @angular/material
```

Choose a prebuilt theme name, or "custom" for a custom theme:

Indigo/Pink

```
? Set up global Angular Material typography styles? No
```

```
? Include the Angular animations module? Do not include
```

Notice the changes to package.json when you add components; this allows an environment to be reproduced by running npm install against the file, similar to the pom.xml file in our Java projects.

Select an example button bank from <https://material.angular.io/components/button/examples> and copy the code to app.component.html to display it

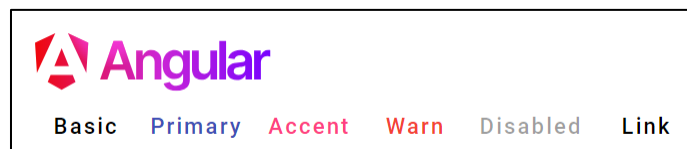
You will need to add the following imports to app.component.ts

```
import { CommonModule } from '@angular/common';
...
import { MatButtonModule } from '@angular/material/button';
...
imports: [CommonModule, RouterOutlet, MatButtonModule],
```

I added my button code to app.component.html, just below the logo:

```
<div class="example-button-row">
  <button mat-button>Basic</button>
  <button mat-button color="primary">Primary</button>
  <button mat-button color="accent">Accent</button>
  <button mat-button color="warn">Warn</button>
  <button mat-button disabled>Disabled</button>
  <a mat-button href="https://www.google.com/" target="_blank">Link</a>
</div>

<h1>Hello, {{ title }}</h1>
<p>Congratulations! Your app is running. 🎉</p>
```



Change the title of the application in app.component.ts:

```
export class AppComponent {
  title = 'Pizza Restaurant';
}
```

Add a data table component to act as a simple list of customer data:

```
ng generate component data-table
CREATE src/app/data-table/data-table.component.html (26 bytes)
CREATE src/app/data-table/data-table.component.spec.ts (641 bytes)
CREATE src/app/data-table/data-table.component.ts (262 bytes)
CREATE src/app/data-table/data-table.component.scss (0 bytes)
```

Create the table and data in the TypeScript component (data-table.component.ts):

```
import { Component } from '@angular/core';
import { MatTableModule } from '@angular/material/table';
```

```

@Component({
  standalone: true,
  selector: 'app-data-table',
  imports: [ MatTableModule ],
  templateUrl: './data-table.component.html',
  styleUrls: ['./data-table.component.scss']
})
export class DataTableComponent {
  displayedColumns: string[] = [
    'userId',
    'lastName',
    'firstName',
    'phoneNumber'];
  dataSource = CUSTOMER_DATA;
}

export interface Customer {
  userId: number;
  lastName: string;
  firstName: string;
  phoneNumber: string;
}

const CUSTOMER_DATA: Customer[] = [
  { userId: 1, lastName: 'Smith', firstName: 'John', phoneNumber: '555-1234' },
  { userId: 2, lastName: 'Doe', firstName: 'Jane', phoneNumber: '555-5678' },
  { userId: 3, lastName: 'Brown', firstName: 'Charlie', phoneNumber: '555-9101' },
  { userId: 4, lastName: 'Johnson', firstName: 'Emily', phoneNumber: '555-1121' },
  { userId: 5, lastName: 'Williams', firstName: 'Anna', phoneNumber: '555-1314' },
];

```

Create the table structure in data-table.component.html:

```

<table mat-table [dataSource]="dataSource" class="mat-elevation-z8">

  <!-- User ID Column -->
  <ng-container matColumnDef="userId">
    <th mat-header-cell *matHeaderCellDef> User ID </th>
    <td mat-cell *matCellDef="let element"> {{element.userId}} </td>
  </ng-container>

```

```

    <!-- Last Name Column -->
    <ng-container matColumnDef="lastName">
      <th mat-header-cell *matHeaderCellDef> Last Name </th>
      <td mat-cell *matCellDef="let element"> {{element.lastName}}
</td>
    </ng-container>

    <!-- First Name Column -->
    <ng-container matColumnDef="firstName">
      <th mat-header-cell *matHeaderCellDef> First Name </th>
      <td mat-cell *matCellDef="let element"> {{element.firstName}}
</td>
    </ng-container>

    <!-- Phone Number Column -->
    <ng-container matColumnDef="phoneNumber">
      <th mat-header-cell *matHeaderCellDef> Phone Number </th>
      <td mat-cell *matCellDef="let element"> {{element.phoneNumber}}
</td>
    </ng-container>

    <tr mat-header-row *matHeaderRowDef="displayedColumns"></tr>
    <tr mat-row *matRowDef="let row; columns: displayedColumns;"></tr>

  </table>

```

Add some styling to the data-table.component.scss file:

```

table {
  width: 100%;
  border-collapse: collapse;

  th.mat-header-cell, td.mat-cell {
    padding: 8px;
    border: 1px solid #e0e0e0;
    text-align: left;
  }

  th.mat-header-cell {
    background-color: #f5f5f5;
  }

  tr.mat-row:nth-child(even) {
    background-color: #f9f9f9;
  }
}

```

Make changes to the app.component.ts file so it imports the data table:

```
import { Component } from '@angular/core';
import { CommonModule } from '@angular/common';
import { RouterOutlet } from '@angular/router';
import { MatButtonModule } from '@angular/material/button';
import { DataTableComponent } from './data-table/data-table.component';

@Component({
  selector: 'app-root',
  standalone: true,
  imports: [
    CommonModule,
    RouterOutlet,
    MatButtonModule,
    DataTableComponent ],
  templateUrl: './app.component.html',
  styleUrls: ['./app.component.scss']
})
export class AppComponent {
  title = 'Pizza Restaurant';
}
```

Add the data table to the app.component.html file:

```
<h1>{{ title }}</h1>
<p>Congratulations! Your app is running. 🎉</p>
<app-data-table></app-data-table>
```